ASEN 3502 Computational Methods
Homework 7

See Gradescope for Due Date

In order to practice for the exam, please do these problems **by hand** (a calculator is fine for the arithmetic); show your work and submit one pdf.

Question 1 (Standard form)
Write the general multivariable constrained optimization problem in standard form (as in class and in Martins & Ning, Ch. 5), with the decision variables collected in a vector $\vec{x}$. Then label each part of what you wrote:

(a) objective function
(b) decision variables
(c) equality constraints
(d) inequality constraints

Also state what it means for a point $\vec{x}$ to be *feasible*, and what it means for an inequality constraint to be *active* at $\vec{x}$.

Question 2 (Fuel tank design and the KKT conditions)
An external fuel tank is to be designed for a wing pylon. At this stage we approximate it as a circular cylinder of radius r and length l (the detailed aerodynamic shape will be designed later). It is made of sheet metal with areal density $s = 10$ kg/m^2 and holds fuel of density $\rho = 800$ kg/m^3 (neglect the thickness of the walls when computing the volume). We want to choose r and l to minimize the total mass of the full tank (fuel plus skin, including the two end caps), subject to the requirements that the tank hold at least $V_{\min} = 1$ m^3 of fuel and that the radius not exceed $r_{\max} = 0.4$ m for ground clearance.

(a) Write the problem in standard form with $\vec{x} = [r,\ l]^{\mathsf{T}}$: give $f(\vec{x})$ and $\vec{g}(\vec{x})$ (there are no equality constraints).

(b) State the KKT conditions for a problem with only inequality constraints, using multipliers σ_j for the g_j.

(c) Two candidate designs are proposed:

$$\text{A: } r = 0.542 \text{ m},\ l = 1.084 \text{ m} \qquad \text{B: } r = 0.4 \text{ m},\ l = 1.989 \text{ m}.$$

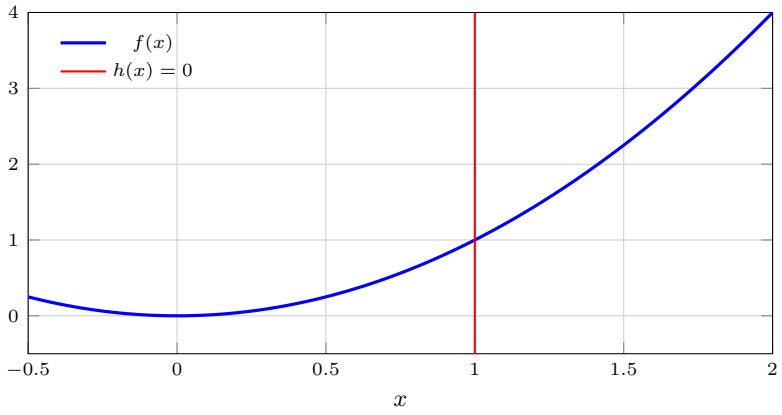
Evaluate $\vec{g}$ at each (small rounding error is fine). Which of these designs is feasible?

(d) For the feasible candidate, compute ∇f and the ∇g_j, then solve for the multipliers σ_1 and σ_2. Does the design satisfy the KKT conditions?

(e) ~~Interpret the multiplier on the radius constraint: if the ground clearance allowed a slightly larger radius, how much mass would be saved per meter of additional allowed radius?~~ (Removed: we did not have time to cover multiplier sensitivity in class.)

Question 3 (Augmented Lagrangian)
Consider the scalar equality-constrained problem (The same f and solution as the constrained optimization demo, but with the constraint written as an equality. The constraint trivially forces the solution to be $x^* = 1$, but the problem is useful for illustrating the augmented Lagrangian method.)

$$\begin{aligned} \underset{x}{\text{minimize}} \quad & f(x) = x^2 \\ \text{subject to} \quad & h(x) = 1 - x = 0. \end{aligned}$$



(a) Find the exact solution x^* and the Lagrange multiplier λ^* by solving the KKT conditions $f'(x) + \lambda\, h'(x) = 0$, $h(x) = 0$.

(b) Write out the augmented Lagrangian

$$\hat{f}(x; \lambda, \mu) = f(x) + \lambda\, h(x) + \frac{\mu}{2}\, h(x)^2$$

for this problem and find $\partial \hat{f} / \partial x$.

(c) In the next two parts, we will complete one iteration of the augmented Lagrangian algorithm, computing x_0^* and x_1^*. Starting from $\lambda_0 = 0$ and $\mu_0 = 2$, find the minimizer x_0^* of $\hat{f}(x; \lambda_0, \mu_0)$ by setting $\partial \hat{f} / \partial x = 0$ (for this problem you can solve it exactly by hand). Evaluate $h(x_0^*)$.

(d) Update the multiplier and penalty parameter with

$$\lambda_1 = \lambda_0 + \mu_0\, h(x_0^*), \qquad \mu_1 = \rho\, \mu_0, \quad \rho = 2,$$

and find the next minimizer x_1^* of $\hat{f}(x; \lambda_1, \mu_1)$ and evaluate $h(x_1^*)$.

(e) Verify that the Lagrange multiplier estimate (λ_1) and minimum estimate (x_1^*) get closer to the true optimum than λ_0 and x_0^*.